Mr. Ahsan | **STA100**

Final Syllabus

1. 10! = 3628800
2. The player well shuffles the pack of 52 playing card, then the probability of a black card from 52 playing cards is: 13/52
3. The probability of drawing a 'jack card 'from 52 playing cards is: 4/2
4. Which dispersion is used to compare variation of two series? Q.D
5. All the values fall on the same straight line and the line has a positive slope then what will be the value of the correlation coefficient 'r': r = +1
6. Regression line Y= a + bX, the value of the correlation coefficient will be zero if: Intercept a = 0
7. When two coins are tossed the probability of at least one head is: 3/4
8. Which one of the following measurement does *not* divide a set of observations into equal parts?
   curve has a longer tail to the right, it is called : standard deviations
9. the model Y = mX + a, Y is also known as the: Predicted variable
10. According to empirical rule approximately 95% of the measurements will fall under which interval? X=2S
11. Which one of the following is written at the top of the table? Title
12. The curve has a longer tail to the right, it is called: Positively skewed
13. Which one of the following is the class frequency? The number of observations in each class
14. If X f(x) is a discrete random variable, then the function: A distribution function
15. Which one of the following graphs is used for a time series data? Histogram
16. If you connect the mid-points of rectangles in a histogram by a series of lines that also touches the x-axis from both ends, what will you gets? Frequency curve
17. If mean of the two observations is 10.5, then median of these two observations will be: 10.5

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18. Which one is the formula of mid range: $x_0 x_m/2$
19. Which one of the following is not included in measures of central tendency: Quartile deviation
20. For the given data 2, 3, 7, 0, -8 G. M will be: Undefined
21. For a particular data the value of Pearson's coefficient of skewness is greater then zero. What will be the shape of distribution? Positively skewed
22. In measures of relative dispersion unit of measurement is: Vanish
23. The F-distribution always ranges from: 0 to $+\infty$
24. In chi-square test of independence the degrees of freedom are: n – p
25. The Chi- Square distribution is continuous distribution ranging from: $0 \leq \chi 2 \leq \infty$
26. If X and Y are random variables, then $E(XY)$ is equal to: $E(X)E(Y)$
27. If $\hat{y}$ is the predicted value for a given x-value and b is the y-intercept then the equation of a regression line for an independent variable x and a dependent variable y is: $\hat{y} = mx + b$, where m = slope
28. The location of the critical region depends upon: Alternative hypothesis
29. The t-Distribution is…….Spread out then the standard normal Distribution. More
30. To find the confidence internal for the ratio of two variances we use: F-Distribution
31. How many percent of values are less then 4th deciles in a symmetric distribution? 40
32. The combined distribution of more than two random variables is: Joint Distribution
33. The degrees of freedom for a T-test with sample size 14 is: 13
34. Which of the falling is true for the binomial distribution b(x: n, p): Mean > Variance
35. What is $m\ f$ in the formula of mode? highest frequency
36. The parameters of the binomial distribution b(x; n, p) are: n & p
37. Which of the following is true for the Poisson distribution: mean = variance
38. If a significance level of 1% is used rather than 5%, the null hypothesis is: Less likely to be rejected
39. The variance of the chi-square distribution is: $2v$
40. The degrees of freedom for a t-test with sample size 10 is: 9
41. The value of χ^2 can never be : Negative
42. The total number of samples when sampling is done with replacement : N^n
43. ANOVA was introduced by : R.A Fisher

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44. The test statistic used in analysis of variance procedure follow the .....distribution.: F

45. For testing of hypothesis about population proportion , we use: Z-test

46. A die is rolled. What is the probability that the number rolled is greater than 2 and even: 1/3 2/6 =1/3

47. The probability of drawing a king of spade from a pack of 52 cards is: 1/52

48. An estimator T is said to be unbiased estimator of ? if E (T) =?

49. From point estimation, we always get: Single value

50. The best unbiased estimator for population variance: Sample variance

51. When c is a constant, then E(c) is: c

52. Var(4X + 5) =_16 Var (X) + 5

53. When f(x) is continuous probability function, then P(X = 1) is: 0

54. The hyper geometric random variable is a(an): Discrete variable

55. From a sample of 200 people were asked whether they like a particular product. Fifty said 'yes' and remain said 'no', assuming 'yes' means a success, which of the following is correct? Sample proportion p=0.25

56. In any data set, what percent of values fall in the interval Median ± *Q.D?* 50 per cent

57. $\sum_{i=1}^{5} X_i - 20 = 0$, *then* $X = 0$

58. The height of a student is 60 inches. This is an example of? Continuous data

59. In Statistics, we have MSE which is abbreviation of? Mean square error

60. The deviation of a distribution from symmetry is called: Skewness

61. If E is an impossible event, then P(E) is: 0

62. If a data set has the even number of observations, the median : Is the average value of the two middle items when all items are arranged in ascending order

63. In testing of hypothesis, we always begin it with assumig that: Null hypothesis is true

64. If a continuous probability distribution has $\beta_2 = 2.14$ then what will be peakedness of the distribution? Platykurtic

65. When each outcome of a sample space has equal chance to occur as any other, the outcomes are called: Equally likely

66. The LSD test is applied only if the null hypothesis is: Rejected

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67. Analysis of variance is a procedure that enables us to test the equality of several: Means

68. If a random variable X denotes the number of heads when three distinct coins are tossed, the X assumed the values: 0,1,2,3

69. If X and Y are independent variables, then E (XY) is: E(X).E(Y)

70. The parameters of the binomial distribution b(x; n, p) are: n & p

71. If P (E) is the probability that an event will occur, which of the following must be false: P(E)= - 1

72. The analysis of variance technique is a method for: Comparing three or more means

73. The continuity correction factor is used when: A continuous distribution is used to approximate a discrete distribution

74. Stem and leaf is more informative when data is : Less than 100

75. The branch of Statistics that is concerned with the procedures and methodology for obtaining valid conclusions is called: Inferential Statistics

76. Which of the following is a systematic arrangement of data into rows and columns: Tabulation

77. In normal distribution Q.D = 0.6745

78. In normal distribution β_2 3

79. If you connect the mid-points of rectangles in a histogram by a series of lines that also touches the x-axis from both ends, what will you get? Frequency curve

80. Which one of the following statements is true regarding a population? It must be a large number of values

81. In a simple linear regression model, if it is assumed that the intercept parameter is equal to zero, then: The slope of the line will also be equal to 0.

82. The degrees of freedom for a t-test with sample size 10 is: 9

83. A failing student is passed by an examiner is an example of: Type II error

84. Which dispersion is calculated from all the observations? Standard deviation

85. Standard deviation of the data 7, 7, 7, 7, 7, 7, 7 is: 0 Standard deviation will always be zero if all the values in data are same

86. Which one is the poor measure of dispersion in open-end distribution? Range

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87. Men tend to marry women who are slightly younger than themselves. Suppose that every man married a woman who was exactly 5 years younger than themselves. Which of the following is correct: The correlation is 1

88. Sum of absolute deviations of the values is least when deviations are taken from: Mean

89. Which of the following measures of central location is affected most by extreme values: Mean

90. Which of the following is a critical value of Z when 1□□ □ 95% for one tailed test: 1.96

91. The difference between expected value of statistic and parameter is called: Bias

92. In Statistics, we have MSE which is abbreviation of: Mean square error

The following data shows the number of hours worked by 200 statistics students.

Number of Hours	Frequency
0 - 9	40
10 - 19	50
20 - 29	70
30 - 39	40

What is its class interval? *10*

93. The F-distribution has …… parameter. Two

94. The degrees of freedom for a t-test with sample size 6 is: 5

95. In testing of hypothesis, we always begin it with assuming that Null hypothesis is true

96. Mean deviation is always: Less or equal to S.D

97. Which formula represents the probability of the complement of event A: 1 - *P* (*A*)

98. Ideally the width of confidence interval should be: 0

99. The probability distribution of a statistic is called the: Sampling distribution

100. Two sets A and B are equal if $A \subseteq B$ and $B \subseteq A$ If two sets A and B are equal we write A = B to designate that relationship.

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101. expression b2 – 4ac under the square root – called the discriminant
102. For a particular data the value of Pearson's coefficient of skewness is greater then zero.
103. What will be the shape of distribution? Positively skewed.
104. The F-distribution always ranges from: 0 to +∞
105. The location of the critical region depends upon: Alternative hypothesis .
106. To find the confidence internal for the ratio of two variances we use: F-Distribution
107. The degrees of freedom for a T-test with sample size 14 is3 .
108. The variance of the chi-square distribution is: v 2.
109. The degrees of freedom for a t-test with sample size 10 is: 9.
110. The value of χ2can never be : Negative.
111. .For testing of hypothesis about population proportion , we use:Z-test.
112. Analysis of variance is a procedure that enables us to test the equality of several:Means.
113. The classical definition of probability assumes: Equally likely evens.
114. MSE stand for mean square error.
115. LSD stand for least significance difference.

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